Math 1610, Honors Calculus, Fall 2026, Worksheet 4

October 5, 2026

Name _____

Please write only your own name in the space above.

1. Suppose g is a bounded function on $(-1, 1)$ — that is, there is some M with $|g(x)| \leq M$ for every $x \in (-1, 1)$. Nothing else is assumed about g.

   Show that

$$\lim_{x \to 0} x\, g(x) = 0.$$